

The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform

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- : The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform
- : Syed Danial Ali Shah, Zeinab Nezami, Maryam Hafeez, Syed Ali Raza Zaidi (University of Leeds)
- : IEEE INFOCOM 2025 Workshop (arXiv:2503.07420)
- : O-RAN, 5G, AI, xApp, RIC, SAC

1.1

AI-RAN Alliance **AI-and-RAN**(AI workload RAN workload), O-RAN NRT-RIC workload 99% RAN 100% . **CAORA**, GPU (MIG) workload **SAC(Soft Actor-Critic)**. RAN

2 (Motivation)

RAN network function **dedicated**. (static).

1. **Off-peak**: RAN
2. **AI workload**: GenAI/edge AI inference workload.

RAN workload	AI workload
latency-sensitive, (non-elastic) carrier ()	(elastic) use case best-effort

(coexistence), RAN AI workload,

3 CAORA

CAORA (Converged AI-and-ORAN Architecture) O-RAN 4.

3.1

- **Layer 1 — External Cloud**: AI/ML workload. edge AI inferencing, generative AI, robotics application, **AI Deployment API** RAN (compute cluster).
- **Layer 2 — Programmable RAN**: O-RAN disaggregation O-RU / O-DU / O-CU.
- **Layer 3 — Controller (NRT-RIC)**: traffic prediction, resource allocation, monitoring custom xApp. **E2 interface** RAN.
- **Layer 4 — Resource Management (E2E Orchestrator)**: compute cluster(GPU, MIG, NIC). SAC agent.

3.2 : Y1 interface

Y1 interface.

- **Monitoring xApp** (Layer 3) bandwidth utilization, system latency, network load RAN KPI.
- radio analytics **Y1 interface** E2E orchestrator(Layer 4).
- E2E orchestrator **Y1 consumer** SAC.

“ $xApp$ = , $E2E\ orchestrator(SAC)$ = ”, SAC agent : (i) (predictive analysis), (ii) (real-time adaptation), (iii) peak RAN (priority management).

RAN network AI workload **multi-purpose system**.

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4.1 Task

task K , task $k \in K$:

$$k = \{\tau, r, p\}$$

- τ : task (RAN / AI)
- $r = [r_i, \forall i = 1, 2, \dots, N]$: (resource type)
- $p \in [0, 1]$: ($>$ /)

$$r(k, t), p(k) .$$

4.2

task k t L_1 norm .

$$\Delta(k, t) = \|r(k, t)\|_1 = \sum_{i=1}^n r_i(k, t)$$

task .

$$\Delta_{\text{total}}(t) = \sum_{k \in K} \Delta(k, t)$$

4.3 (System Dynamics)

workload $W(t)$ task , (contention) .

$$W(t) = \int_0^t \left(\sum_{k \in K} P(k, t) \Delta(k, t) - \sum_{k \in K} P_c(k, t) \Delta(k, t) C(k, t) \right) dt$$

- $P(k, t)$: task
- $P_c(k, t) = \alpha \cdot p(k)$: task (α , $p(k)$)
- $C(k, t)$: task k

$$, () .$$

4.4

RAN AI , .

$$r_{\text{RAN}}(t) + r_{\text{AI}}(t) \leq \mathcal{R}_{\text{max}}$$

$\mathcal{R}_{\text{max}}$, NVIDIA MIG ().

5 SAC

E2E orchestrator **SAC(Soft Actor-Critic)** . SAC entropy - , .

- **State:** monitoring xApp RAN/AI (bandwidth utilization, system latency, network load).
- **Action:** RAN AI ().
- **Reward:** task , (utilization) . peak RAN .
“RAN ” “AI workload best-effort ” .

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6.1

- **Python** (O-RAN).
- **NVIDIA MIG(Multi-Instance GPU)** . “ ” GPU instance .
- RAN/AI dynamic scenario .

6.2

- RAN workload **99%** — RAN .
 - RAN **100%** — off-peak AI workload .
 - Peak RAN SAC AI RAN → priority-weighted reward .
 - AI workload RAN , .
- , RAN .

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() , .

1. : RAN (L1/MAC) , MIG . dataset .
2. : task MIG → fractional/variable allocation (PRB/TB) .
3. **NRT-RIC** : Y1/E2 **10 ms~** . “ ” sub-TTI .
4. **AI workload** : AI task (token, attention) “ MIG ” .

8 Gap

AI-RAN **AI-and-RAN(/)** . importance-aware MAC scheduling
AI-for-RAN() , •granularity .

8.1

(CAORA)		
	NRT-RIC xApp + E2E orchestrator NRT-RIC (10 ms~), Y1/E2	GPU-collocated L2 MAC scheduler sub-TTI ,

(CAORA)		
AI workload	MIG (GPU) (MIG)	PRB / TB () token importance (entropy · attention) scheduler utility
“AI-and-RAN”	GPU RAN+AI compute , Python	offloaded VLM RAN importance-aware end-to-end task utility per resource Sionna → cuMAC on GH200, ROS2

8.2

- **AI workload** : AI workload *GPU* . use case multi-robot VLM offloading *uplink*
“AI RAN GPU ” , “AI RAN ”
- : NRT-RIC/Y1 , sub-TTI externalize (GPU-collocated MAC
)
- : monitoring xApp , • **token** **proactive** **PRB**

8.3 Positioning

- predictive AI-and-RAN orchestration , **convergence backbone** , **macro**
- (arXiv:2507.09124, *Proactive AI-and-RAN Workload Orchestration*) , reactive/ proactive
gap anchoring .

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“ *RAN AI GPU(MIG)* ” NRT-RIC + SAC , AI-and-RAN
: RAN (VLM offloading) **token** **sub-TTI MAC** **importance-aware**
— (micro) .